Chpt 15:

(27) $m_e = 9.11 \times 10^{-31}$ kg $\qquad e = 1.602 \times 10^{-19}$ C

$m_p = 1.67 \times 10^{-27}$ kg $\qquad$ (charge of proton)

$\rightarrow$ In 1 kg $= M$,

n_e = number of electrons

$= \dfrac{M}{m_e}$

$\approx 1.1 \times 10^{30}$ electrons

n_p = number of protons

$= \dfrac{M}{m_p}$

$\approx 6.0 \times 10^{26}$ protons

Thus, $Q_e = -n_e e = \boxed{-1.76 \times 10^{11}\ C}$

$Q_p = n_p e = \boxed{9.59 \times 10^{7}\ C}$

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(29) $M = 1$ kg, Carbon: (6p, 6n), 6 electron

$m_c = 12.01\ u$ (unified mass units)

$1\ u = 1.661 \times 10^{-27}$ kg

n_c = number of carbon atoms

$= \dfrac{M}{m_c \cdot u}$

$\approx 5 \times 10^{25}$

$n_e = 6\, n_c \approx \boxed{3 \times 10^{26}}$ [approximate in part because a block of Carbon can have different isotopes like C^{12} and C^{14}]

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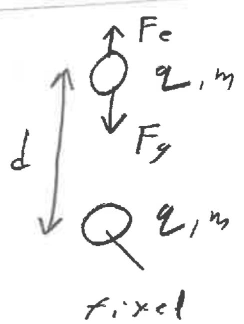
(37) 

$m = 1.4\,g = 1.4 \times 10^{-3}$ kg

$q = 0.450\,\mu C = 0.45 \times 10^{-6}$ C

want to balance weight with electrostatic force

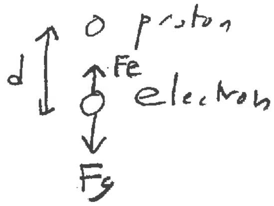
$F_e = \dfrac{k|q|^2}{d^2}$, $F_g = mg$ $\quad (g = 9.8\ m/s^2)$

$\left(k = 8.99 \times 10^9\ \dfrac{N \cdot m^2}{C^2}\right)$

Thus, $\dfrac{k|q|^2}{d^2} = mg \rightarrow d^2 = \dfrac{k|q|^2}{mg}$

$\rightarrow d = \sqrt{\dfrac{k}{mg}}\,|q| = \boxed{0.36\ m} = 36\ cm$

39.



$F_e = F_g \rightarrow \dfrac{k e^2}{d^2} = m_e g$

$\rightarrow d = \sqrt{\dfrac{k}{m_e g}}\, e$ (similar to prob 37)

$= \boxed{5.08\ m}$

$m_e = 9.11 \times 10^{-31}$ kg

$e = 1.602 \times 10^{-19}$ C

$g = 9.8\ m/s^2$

$k = 8.99 \times 10^9\ \dfrac{N \cdot m^2}{C^2}$

Note: since distance within an atom are $\approx 10^{-10}$ m, the above result suggests that the electrostatic force >> gravitational attraction. So gravity can be ignored in atomic calculations.

43.

$q_1 = -25\ \mu C$

$q_2 = -75\ \mu C$

$d = 1$ m

☆ since both q_1 and q_2 are negative, both q_1 and q_2 attract Q if $Q > 0$. If $Q < 0$, then q_1 and q_2 repel Q.

Equate the Force on Q due to q_1 to that due to q_2

$F_{1Q} = \dfrac{k|q_1||Q|}{x^2}$, $F_{2Q} = \dfrac{k|q_2||Q|}{(d-x)^2}$ (both attractive or repulsive)

The $F_{1Q} = F_{2Q} \rightarrow \dfrac{\cancel{k}|q_1|\cancel{|Q|}}{x^2} = \dfrac{\cancel{k}|q_2|\cancel{|Q|}}{(d-x)^2}$

$\dfrac{|q_1|}{|q_2|} = \dfrac{x^2}{(d-x)^2}$

NOTE: charge Q cancels

Denote $\dfrac{|q_1|}{|q_2|}$ by r $\left(r = \frac{1}{3}\right)$

Then $r = \dfrac{x^2}{(d-x)^2} \rightarrow (d-x)^2\, r = x^2$

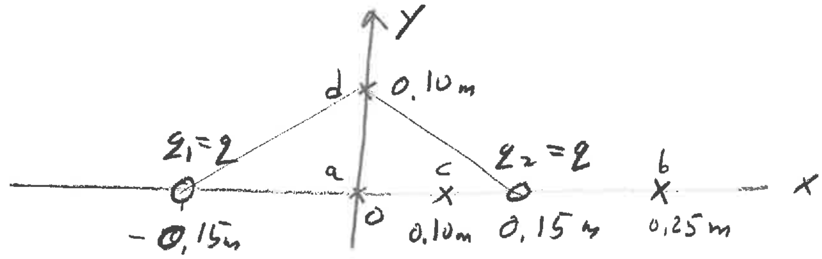
$(d^2 + x^2 - 2dx) = \dfrac{x^2}{r}$

$x^2\left(1 - \frac{1}{r}\right) - 2dx + d^2 = 0$

solve quadratic equation: $ax^2 + bx + c = 0$

$x = \dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \dfrac{2d \pm \sqrt{4d^2 - 4\left(1-\frac{1}{r}\right)d^2}}{2\left(1-\frac{1}{r}\right)} = \begin{cases} \cancel{-1.37\ m} \\ \boxed{0.366\ m} \end{cases}$

55.



$x_1 = -0.15\,m$, $y_1 = 0$
$x_2 = 0.15\,m$, $y_2 = 0$
$q_1 = q_2 = q = 15\,\mu C$

(a) At $(0,0)$:

$$E_1 = \frac{kq_1}{r_1^2} = \frac{kq}{x_1^2} \quad \text{(to right)}$$

$$E_2 = \frac{kq_2}{r_2^2} = \frac{kq}{x_2^2} \quad \text{(to left)}$$

so $\vec{E}_{net} = \vec{E}_1 + \vec{E}_2 = \boxed{0}$ (since $x_1^2 = x_2^2$)

(b) At $(0.25\,m, 0)$:

$$E_1 = \frac{kq}{(0.4\,m)^2} \quad \text{(to right)}$$

$$E_2 = \frac{kq}{(0.1\,m)^2} \quad \text{(to right)}$$

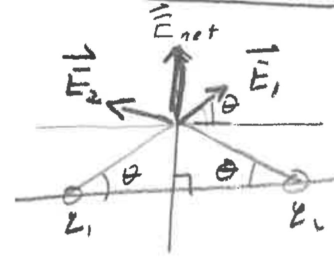
so $$\vec{E}_{net} = kq\left[\frac{1}{(0.4\,m)^2} + \frac{1}{(0.1\,m)^2}\right]\hat{i} \;\boxed{= 106\,kq\,\hat{i}}$$

(c) At $(0.10\,m, 0)$:

$$E_1 = \frac{kq}{(0.25\,m)^2} \quad \text{(to right)}$$

$$E_2 = \frac{kq}{(0.05\,m)^2} \quad \text{(to left)}$$

so $$\vec{E}_{net} = kq\left[\frac{1}{(0.25\,m)^2} - \frac{1}{(0.05\,m)^2}\right]\hat{i} = \boxed{-384\,kq\,\hat{i}}$$



(d) At $(0, 0.10\,m)$:

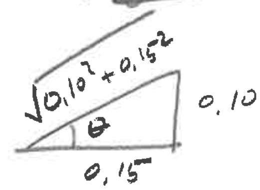
$$E_1 = \frac{kq}{(0.15\,m)^2 + (0.1\,m)^2}$$

$$E_2 = \frac{kq}{(0.15\,m)^2 + (0.1\,m)^2}$$

$$\tan\theta = \frac{0.10\,m}{0.15\,m}$$

Note: magnitudes $E_1 = E_2 \equiv E$

Thus, $\vec{E}_1 = E\cos\theta\,\hat{i} + E\sin\theta\,\hat{j}$
$\vec{E}_2 = -E\cos\theta\,\hat{i} + E\sin\theta\,\hat{j}$



$$\rightarrow \vec{E}_{net} = 2E\sin\theta\,\hat{j}$$

$$= \frac{2\,kq \times 0.10\,m}{\left((0.15\,m)^2 + (0.1\,m)^2\right)^{3/2}}\,\hat{j} = \boxed{34\,kq\,\hat{j}}$$

$$\sin\theta = \frac{0.10}{\sqrt{0.10^2 + 0.15^2}}$$

so $\boxed{(a), (d), (b), (c)}$ is ranking of magnitudes from smallest to largest

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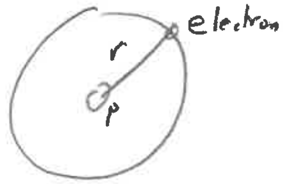


$r = 5.29\times10^{-11}\ m$

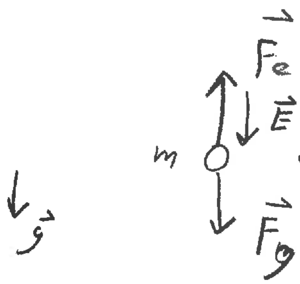
a) $E = \dfrac{kq_p}{r^2}$ where $q_p = e$ (= charge of proton)

$= \boxed{5.15\times10^{11}\ \frac{N}{C}}$ (radially away from proton)

b) $F_e = q_e E = \dfrac{k q_e q_p}{r^2}$ (radially toward proton)

$= \boxed{8.25\times10^{-8}\ N}$

75



oil drop ($q = -3e$)

$E = 11.5\times10^{3}\ N/C$

$\rho = 940\ kg/m^3$
= density of oil droplet

$m = \rho \cdot \frac{4}{3}\pi r^3$

a) $\vec{E}$ must point downward since $q < 0$ and $\vec{F_e} = q\vec{E}$ balances gravity $\vec{F_g} = m\vec{g}$ (downward)

b) $|q|E = mg = \rho \frac{4}{3}\pi r^3 g$

solve for r: $\dfrac{3|q|E}{4\pi\rho g} = r^3$

$\rightarrow r = \left(\dfrac{3|q|E}{4\pi\rho g}\right)^{1/3} = \boxed{5.23\times10^{-7}\ m} = 523\ nm$